

Shreyas Deb

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EDUCATION

International Institute of Information Technology, Hyderabad

Aug 2023 – May 2028

B.Tech. in Computer Science & M.S. by Research in Computational Natural Sciences (CGPA: 8.85) Hyderabad, Telangana

EXPERIENCE

Undergraduate Researcher

Aug. 2025 – Present

Center for Computational Natural Sciences and Bioinformatics (CCNSB), IIIT Hyderabad

Hyderabad, Telangana

- Conducting research under **Dr. Vinod P K**, focusing on computational modeling of complex biological systems through the lens of **Systems and Network Biology**.
- Investigating the intersection of **Deep Learning** and **Network Biology** to decipher emergent disease mechanisms, integrate multi-omics data, and predict clinical outcomes for **precision medicine**.
- Built and evaluated **deep learning architectures** for high-dimensional **multi-omics data integration**, emphasising model design, experimental rigour, and interpretable prediction of clinical outcomes.
- Reimplemented **5+ ML/DL pipelines** for noisy, heterogeneous biological data, focusing on quantitative modelling and representation learning.

AI/ML Engineering Intern

May 2025 – Jul. 2025

MontyCloud

Bengaluru, Karnataka

- Built a production-grade **Agentic Suite** for Product Management as part of the **PM360 team**, including a **Spec Writer**, **Meeting Analyzer**, and **Product Value Agent**, increasing PM efficiency by **40%**.
- Designed **multi-agent LLM systems** using **GraphRAG** pipelines grounded in a **Neo4j Knowledge Graph** with temporal memory and vector search (via **Azure AI Search**), deployed on a fully **serverless AI backend** using **AWS**, with agent workflow orchestration via **AWS Strands** and **MCP (Model Context Protocol)** servers.
- Engineered data ingestion and retrieval pipelines from **Jira** and **Confluence**, and delivered a fault-tolerant, high-availability system deployed across **US, EU, and India**.
- Achieved **80%+ reduction** in spec creation time and **90%+ spec accuracy**; delivered robust ingestion pipelines from Jira, Confluence, and meeting transcripts.

Software Engineering Intern

Jan. 2025 – Apr. 2025

uExcelerate

Hyderabad, Telangana

- Developed a scalable **Learning Management System (LMS)** with an enhanced UI and personalized content delivery, increasing user engagement by **35%** and improving average session duration by **25%**.
- Designed and implemented a modular **recommendation pipeline** using hybrid ranking (content-based, collaborative, and interest-based), achieving **80%+ accuracy** with clean extension points for future personalization features.

Front-end Development Intern

Jan. 2025 – Feb. 2025

Texperia

Bengaluru, Karnataka

- Optimized the frontend architecture for the **Playda platform**, achieving a **32% reduction** in average page load time and improving interaction responsiveness by **~25%** for monthly visitors.
- Contributed to the **U.S. market expansion initiative** by redesigning and localizing U.S.-facing pages, directly supporting a projected **40% increase** in user acquisition at launch.

PROJECTS

PairsTrader – Statistical Arbitrage Engine | *Python, Mathematical Modelling, Statistics, Finance*

[GitHub](#)

- Developed a modular **stat-arb engine** for NSE equities using **Engle-Granger cointegration** for pair selection and **PCA + DBSCAN clustering** to identify correlated pairs from price dynamics.
- Implemented **Kalman Filter** for time-varying hedge ratios (adaptive beta + dynamic intercept), preserving spread stationarity across regimes – **40% Sharpe improvement** over static OLS on validation pairs.
- Built **15+ mean-reversion features** (AR(1) half-life, momentum, volatility, correlation) for XGBoost classifier; backtests revealed baseline Z-score outperformed ML gating, highlighting simplicity's value in low-frequency trading.

StockSeqCompare – Stock Forecasting & Trading Signals | *Python, Statistical Alpha, Finance*

[GitHub](#)

- Developed a complete pipeline predicting next-day log returns using ensemble of sequence models (**CNN-LSTM, BiLSTM+Attention, Transformer**); CNN-LSTM delivered **57.7% directional accuracy** and annualized **Sharpe ratio of 2.91** on 5-year AAPL data.
- Created **10+ technical indicators** (MACD, RSI, volatility, momentum) with expanding-window z-score normalization to eliminate look-ahead bias in non-stationary price series.
- Backtested long/short trading strategy generating **+53% total return** (2× buy-and-hold) while limiting maximum drawdown to **-8.5%**.

Docs++ – Distributed File System | *C, Networking, Concurrency, POSIX, System Design*

[GitHub](#)

- Built a distributed document system from scratch in **C** with **sentence-level locking** for concurrent multi-user editing, real-time access control, and persistent versioned storage.
- Architected a **3-tier distributed system** with a Name Server, multiple Storage Servers, and clients, supporting **12+ filesystem operations** including streaming, hierarchical namespaces, versioning, and ACL-based permissions.
- Implemented checkpoint-based version control, undo support, persistence, and efficient file lookup with **zero external libraries**.

AIshA – Intelligent Unix Shell | *C99, POSIX APIs, termios, OpenSSL, Signal Handling*

[GitHub](#)

- Built a Unix shell from scratch in **C99** with **40+ built-in commands**, background job control, and natural-language command translation via **Gemini AI** over a hand-rolled HTTPS client.
- Implemented raw **termios-based line editing** with autocomplete and history navigation; wrote a custom JSON parser for AI responses – **zero external runtime dependencies** beyond OpenSSL.
- Designed POSIX-compliant process and signal handling around **fork/exec/waitpid** and **SIGINT/SIGTSTP/SIGCHLD**, with a modular command-dispatch architecture mapping tokens to handler functions.

Saathi – AI Mental Wellness Platform | *Next.js, FastAPI, OpenAI, Multi-Agent Systems*

[GitHub](#)

- **HackIIIT 2025 Most Impactful** winner; built a full-stack **PWA** with **5 specialized AI agents** for active listening, guided coping, crisis escalation, privacy enforcement, and multi-disciplinary advisory.
- Designed a safety-aware **multi-agent orchestration layer** in FastAPI with shared context, fallback chains, and hard safety boundaries; complete Next.js PWA delivered end-to-end in **under 24 hours**.
- Implemented **gamification features** (achievement badges, mood-based Spotify integration, dynamic color schemes, crisis mode for immediate escalation) and **privacy-first journaling** with AI-powered insights, anonymous community forums, and guided reflection prompts.

shAI – Natural Language Shell Assistant | *Python, LLMs, Ollama, CLI Tools*

[GitHub](#)

- Created a **CLI tool** that translates natural language to bash commands using **local Ollama models (CodeLlama)**, with query validation and safety checks before execution.
- Implemented **3-stage validation pipeline**: query appropriateness check, command safety analysis, and detailed execution reporting with error handling.
- Achieved **local-first privacy** by running entirely on-device without cloud dependencies, suitable for security-conscious environments.

ACHIEVEMENTS

Dean's List & Merit List – Recognition for exceptional academic performance (top 15% of the batch) at IIIT Hyderabad (2023–2024).

HackIIIT 2025 Most Impactful – Won Most Impactful award at HackIIIT 2025 for Saathi, an AI mental wellness platform.

Action Plan Finalist – Selected as **Top 25 finalist** (from 3000+ entries) at Action Plan, the flagship social entrepreneurship event of Abhyuday — The Social Body of IIT Bombay's Advitya Social Fest.

TECHNICAL SKILLS

Languages: C, C++, Python, Rust, Go, JavaScript/TypeScript, SQL, Bash, HTML/CSS, x86/RISC-V Assembly

Systems & Low-level: POSIX APIs, Socket Programming, Multithreading & Synchronization, Lock-free Data Structures, Memory Management, TCP/IP, Linux Kernel, System Design

Backend: FastAPI, Flask, Node.js, Express.js, GraphQL, REST, Supabase

Frontend: React, Next.js, Vite, Svelte, Tailwind CSS, Progressive Web Applications

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Neo4j

AI/ML & Data: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, LangGraph, OpenAI API, Ollama, RAG (GraphRAG), MCP, Vector DBs (Pinecone, ChromaDB, Azure AI Search)

Cloud & DevOps: AWS (Lambda, Strands, S3, DynamoDB), Azure, Docker, Kubernetes, Git, GitHub Actions, Jira, Confluence, Bitbucket

RELEVANT COURSEWORK

Data Structures & Algorithms | Operating Systems & Networks | Computer Systems Organization | Distributed Systems | Algorithm Analysis and Design | Design & Analysis of Software Systems | Statistical Methods in AI | Probability & Random Processes | Linear Algebra | Discrete Mathematics | Pricing Options with Mathematical Models (Coursera)

Tech Team Head

Aug. 2024 – Present

E-Cell IIIT-Hyderabad

- Directed all technical operations for the Entrepreneurship Cell, maintaining server infrastructure at **98% uptime**.
- Led the engineering team for **Megathon 2025**, Hyderabad's largest student-run hackathon (**1,000+ participants**), overseeing registration portals and live event dashboards.
- Assisted in organizing flagship events and workshops, fostering a spirit of entrepreneurship and innovation across the campus community.

Project Manager & Maintainer

Mar. 2025 – Present

Open Source Developers Group (OSDG), IIIT-Hyderabad

- Maintain **Whispr**, a privacy-first open-source review platform serving **1,000+ students** across **200+ courses** and **100+ professors**.
- Mentor open-source contributors, run technical workshops, and steward student-run repositories to grow the campus open-source ecosystem.
- Drive contribution culture by onboarding new developers, establishing code review standards, and organizing contributor sprints.